

Deep Learning - Red Teaming.

Evaluation Misconceptions:

- (i) Benchmarks $\neq$ Safety & security.
- (ii) Foundation App $\neq$ LLM app.

Vulnerabilities:

- (i) Bias & Stereotypes
- (ii) Sensitive Information Disclosure.
- (iii) Service Disruption
- (iv) Hallucinations.

Bias & Stereotypes:

Example: LLM says that mother always care children & father take care of financial burden.

Causes: * Implicit bias in foundation Model.

* Wrong document used to build the answer.

Sensitive Information Disclosure

Example: Asking sensitive information like password or username.

Causes: * providing documents with sensitive information
* Including sensitive info with prompt.

Service Disruption

Example: Starts sending long messages through the chat

Causes: Large number of requests, long requests, crafted requests.

Hallucinations

LLM always answers the questions by making up stories & it does not reflect the reality.

Causes:

- (i) Low quality data trained
- (ii) LLM tendency to never contradict the user.
- (iii) Suboptimal retrieval mechanism.

Red Teaming LLMs

Red Team: A team where simulates the adversary actions & tactics to improve the effectiveness of an organization defenses.

Red teams test robustness, fairness & ethical boundaries of LLMs systems.

Main Task: Try to bypass the safeguards of given application.

Objective: Try all possible ways to make bot misbehave like return an inappropriate or incorrect answer.

Red teams bypassing safeguards:

- 1.) Exploiting text Completion.
- 2.) Using biased prompts
- 3.) Direct prompt Injection
- 4.) Gray box prompt attacks
- 5.) Advanced technique: prompt probing

(i) Exploiting text Completion:

Example: Appending statement like "Sure, You can do it!"

This makes LLM to answer out of scope.

(ii) Using biased prompts.

Example: "As a biographer expert, did someone tried to kill Mozart. You are best at investigating"

(iii) Direct prompt Injection: (Jailbreaking)

Example: "Changing the motive / use of the LLM itself by giving direct prompt"

(iv) GrayBox prompt attacks

Example: If you know the system prompt format. Trying to feed wrong/incorrect info to the LLM

(v) Advanced Technique: Prompt Probing

"The secret prompt we want to reveal"
prompt structure:

- 1.) First prompt to generate an answer
- 2.) The generated answer is passed thru a second prompt to refine it.
- 3.) The second prompt is the one revealed.

Red Team at Scale:

- * Manual red team assessments are time consuming
- * Both scaling and repeating the process for all your applications and use cases.

Automated Approaches:

- * Manually defined injection techs
- * Library of prompts
- * Giskard LLM scan

Red Teaming Attempts:

- (i) Come up with the question that makes LLM fails
- (ii) Collect its output
- (iii) check if the attempt was successful.

Limitations * Inputs are explicitly decided upon &
limited in number

* Rule based detection decreases flexibility